

Service Engineering Application

1. Service Candidate Identification

In the transition to a Service-Oriented Architecture (SOA), we have identified two high-value service candidates that provide the most utility to our School Management System:

Candidate A: Academic Grade Service

What it does?

The Academic Grade Service is the primary engine for managing student performance data. It acts as a digital ledger where instructors can securely record assessment results, midterms, and final exam marks. Instead of having grades scattered across different spreadsheets, this service centralizes everything into one secure location.

- **Justification for Selection: * Reusability across processes:** This service is the most heavily reused component in our ecosystem. It serves the Student Portal for grade viewing, the Teacher Console for grade entry, and the Administrative Module for generating official transcripts.
 - **Independence & External Potential:** The service is designed to be completely decoupled from our specific database technology. It has high potential for external usage, such as allowing a centralized national database to verify student results through a secure connection.
- **Service Classification: Business Service.** It implements core school logic rather than general-purpose technical tasks.

How it works in the system

1-For Teachers: It provides a simple way to "POST" new grades or "PUT" updates if a student's score needs correction after a regrade.

2-For Students: It allows them to "GET" their specific results instantly through the student portal as soon as they are published.

3-Key Advantage: It is a Business Service because it handles the school's most sensitive data—academic records—and ensures that only authorized staff can make changes, which supports our overall Security goals.

Candidate B: Attendance & Presence Service

What it does?

The Attendance & Presence Service is a dedicated tool for tracking student engagement and physical presence in the classroom. It moves beyond simple paper rosters by creating a real-time record of who is in class, which is vital for both academic success and school safety.

- **Justification for Selection:**
 - **Reusability across processes:** Attendance data is required by multiple departments. For example, the Exam Eligibility module uses it to check if a student met the attendance threshold, and the Security Module uses it to verify who is currently on campus.
 - **Independence & External Potential:** It is an independent "plug-and-play" module. This allows us to integrate it with external hardware, such as student ID card scanners or mobile apps, without needing to modify the rest of the software.
- **Service Classification: Business Service.**

How it works in the system

1-Daily Tracking: Every morning or at the start of a lecture, the system creates a new entry for each student.

2-Calculations: The service doesn't just store dates; it automatically calculates a student's total attendance percentage. For example, if a student misses too many classes, this service flags them as "at risk."

3-Key Advantage: This is also a Business Service because tracking attendance is a core requirement of the educational process. It is highly Reusable because it can be connected to mobile apps or future hardware, like card scanners, without needing to change the main software.

2. Service Requirements Specification

A) Grade Service Specification

- **Functional Requirements:** The service must allow teachers to enter and update marks and allow students to view their specific academic transcripts.
- **Non-Functional Requirements:**
 - **Security:** Only authorized users with "Teacher" or "Admin" status can modify grade data.
 - **Performance:** The service must be optimized to handle hundreds of students checking results at the same time without slowing down.
- **Inputs/Outputs & Constraints:**
 - **Inputs:** Student ID, Course Name, and the numeric Grade.
 - **Outputs:** A standardized digital grade report.
 - **Constraints:** Once a semester is officially closed, grades can no longer be modified through the service.

B) Attendance Service Specification

- **Functional Requirements:** The service must log daily presence and provide cumulative totals for every student.
- **Non-Functional Requirements:**
 - **Availability:** The service must be available 24/7 to ensure morning check-ins and late-night reports can be processed without interruption.
- **Inputs/Outputs & Constraints:**
 - **Inputs:** Student ID, Date, and Presence Status.
 - **Outputs:** An attendance summary for a specific student or class.
 - **Constraints:** A student cannot be marked "Present" more than once for the same class on the same day.

3. Service Interface Design (The Roadmap)

I have designed these services as RESTful APIs:

- **Endpoints & HTTP Mapping:**
 - **GET /api/grades/{studentID}:** Used by Sayed to retrieve a student's history.
 - **POST /api/grades:** Used by Sayed to create a new record.
 - **PUT /api/attendance/{id}:** Used by Hany to update an existing status.
 - **DELETE /api/attendance/{id}:** Used by Hany to remove a mistake or duplicate entry.
 - **Data Formats:** All communication will be handled using JSON, ensuring that the data is easy to read and compatible with any device.
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4. Service Composition Workflow

- **Service Name:** "Final Student Evaluation Workflow".
- **Workflow Explanation:** 1. The workflow begins by calling the Attendance Service to see if a student has enough presence to be eligible for final grades. 2. If eligible, it then automatically calls the Grade Service to pull all their scores for the term. 3. Finally, it combines this information into one final "Completion Report" that is sent to the student.